

Jumping-At-Air : Jumping without Touching the Ground in Virtual Reality

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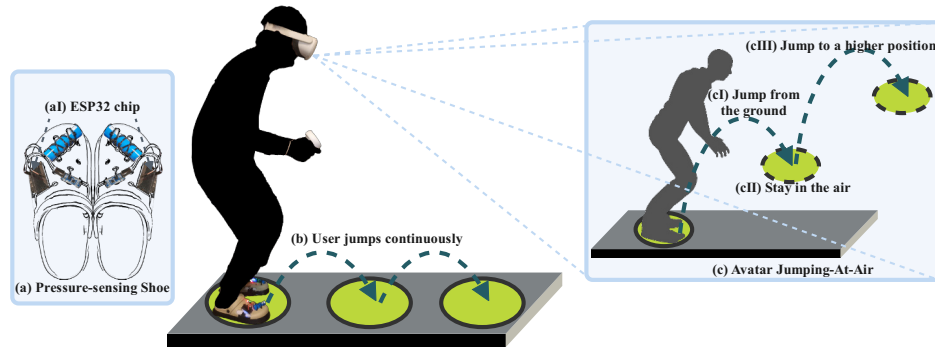


Figure 1: Pressure-sensing shoes(a) are a shoe sole pressure collection device based on the ESP32 chip(ai), which can accurately determine the user's jumping stage. Based on this, the Jumping-At-Air(c), simulating the Double Jumping mechanism in video games, was implemented. When the user jumps continuously(b), the virtual avatar can jump from the ground(cii), stay in the air(cii), and jump to a higher position then(ciii).

ABSTRACT

We propose Jumping-At-Air, where the avatar jumps in the virtual world and lands at a higher air while the user jumps in the real world, then jumps again from the air. The core idea is to first create a pair of pressure-sensing shoes based on the ESP32 chip, equipped with pressure sensors in the soles, to accurately detect the jumping stage. Then, through a carefully designed equation with two adjustable parameters, the user's jump trajectory in the real world is redirected to drive the jumps of the avatar in the virtual world. The redirected jumping curve is differentiable and continuous throughout its domain and conforms to the definition of a parabola.

Index Terms: Human-centered computing, Human computer interaction (HCI), Interaction paradigms, Virtual Reality.

1 INTRODUCTION

The core idea of the redirection jumping technique is to decompose virtual reality jumps into five consecutive stages (see Figure 2): *Standing*, *Ready*, *Ascending*, *Descending*, and *Landing* [2]. Then, during the *Ascending* and *Descending* stages, the jump curve is adjusted using a gain factor, making the avatar's jumping in the virtual world different from the user's jumping in the real world. It can allow the avatar to jump to a higher position [3, 4] in the virtual world and the user will not be aware of this.

In this paper, our main contributions are as follows.

1. **A high-precision, low-cost jump detection algorithm based on pressure sensor shoes.** We designed an accurate jump detection device in the form of specially designed shoes equipped with pressure sensors in the soles, provides a fast and highly accurate way to detect jump actions.

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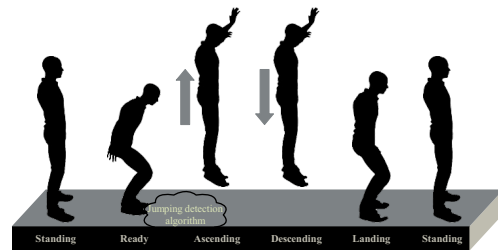


Figure 2: The fundamental framework of redirected jumping consists of five stages: *Standing*, *Ready*, *Ascending*, *Descending*, *Landing*, *Standing*. A jumping detection algorithm is used to determine the start and end of a user's jump

2. **The Jumping-At-Air that offers surreal and efficient vertical movement.** We introduced a jumping form inspired by the widely used *Double Jumping* [1] game mechanic in games into redirection jumping technology, naming it *Jumping-At-Air*.

2 METHODOLOGY

All of our innovative work can be divided into two parts: one is the precise and efficient jump detect algorithm, and the other part is the novel redirect jumping form Jumping-At-Air.

2.1 Jump Detection

We have created a pair of pressure-sensing shoes (Figure 3) and designed a jumping detection algorithm based on this.

As mentioned above, the jump consists of five consecutive stages (Figure 2), and a transition condition needs to be set between each pair of adjacent stages. Once this condition is met, it is considered to have entered the next stage (Figure 4).

2.2 Jumping-At-Air

Jumping-At-Air is inspired by the common Double Jumping[1] mechanism in games, where a game character can jump from the

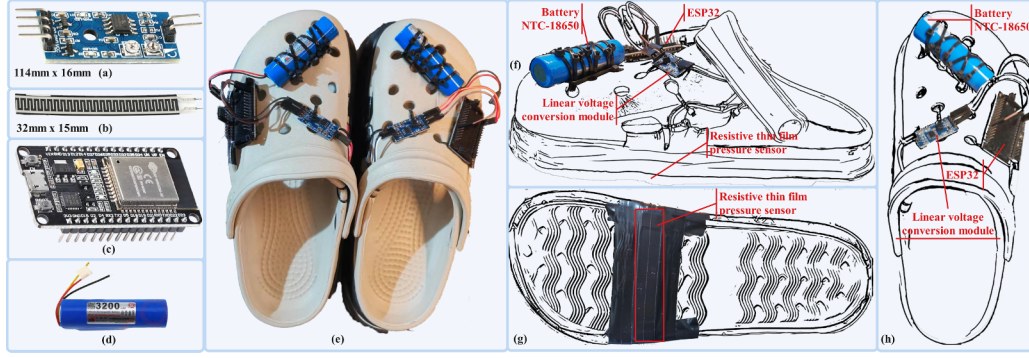


Figure 3: Schematic diagram of the pressure-sensing shoes: (a) is the Linear voltage conversion module, (b) is the Resistive thin film pressure sensor, (c) is the ESP32 chip, (d) is the NTC-18650 Battery, (e) is a real photo of the shoes, (f) is the side view of the shoe, (g) is the bottom of the shoe, and (h) is the top view of the shoe

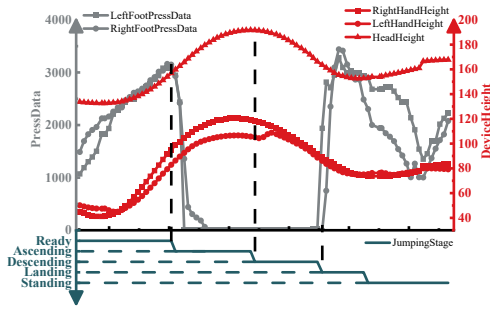


Figure 4: The precise timing of the transition in the jumping stage can be ascertained with accuracy through the utilization of pressure data and devices' height.

ground and jump again in the air to reach higher places. To achieve Jumping-At-Air, we have designed an innovative redirected jumping method.

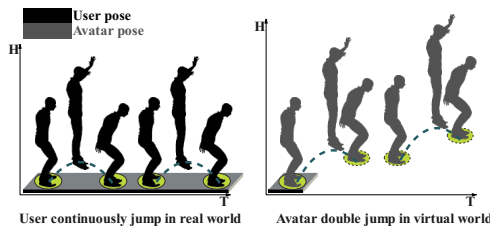


Figure 5: Core idea of Jumping-At-Air.

Let h_r represent the user's height in the real-world setting, v_{0r} denote the initial velocity of the user's jumping in the real world, and g_r indicate the gravitational acceleration experienced by the user in the real world. Correspondingly, in the virtual world, the height of the avatar is denoted as h_v , the initial velocity of the avatar's jumping is represented as v_{0v} , and the gravitational acceleration experienced by the avatar is denoted as g_v . The parameter t represents the elapsed time. Thus, the relationship between these variables can be described by the following equations:

$$h_v = 2kv_{0r}t - \sqrt{2kg_r h_d}t - \frac{1}{2}kg_r t^2 \quad (1)$$

$$h_v = 2kv_{0r}t - \sqrt{2kv_{0r}g_r h_d}t - \frac{1}{2}kv_{0r}g_r t^2 \quad (2)$$

By employing the above values, we can compute the avatar's jumping trajectory in the virtual scene.

We refer to the Jumping-At-Air driven by Equation 1 as Jumping-At-Air V1, and the one driven by Equation 2 as Jumping-At-Air V2.

3 RESULTS

The results indicated that traditional jump detection algorithms can lead to false detection and missed detection. Even when the traditional method operates under its optimal parameters (with a detection threshold of $0.04m$), our jump detection algorithm based on pressure sensors demonstrates significant advantages in both accuracy and speed.

Although Jumping-At-Air represents a surreal jump, users generally exhibit high acceptance. After brief acclimation, they can effectively control their avatar to reach desired heights using Jumping-At-Air.

4 CONCLUSION

First, we propose a jump detection algorithm based on pressure-sensing shoes, which significantly improves the accuracy and precision of jump detection through the use of an affordable shoe-like hardware. Next, we introduce a new form of redirected jumping, inspired by the Double Jumping mechanism in games, allowing users to control avatars for surreal jumps, enabling more efficient vertical movement. Experiments show that this form of jumping has good controllability and provides a comfortable user experience.

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